

Exact Scale Covariance Forces Degenerate Translation-Invariant Landmark Kernels

Run `fa_banach_001`, lane 12

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Status

This packet gives a sharp classification theorem for a precise subproblem suggested by Micheli and Glaunès [1]. No nondegenerate fixed, continuously differentiable, translation-invariant positive-definite matrix kernel can make landmark Hamiltonian dynamics exactly covariant under all spatial dilations. The obstruction persists even if each dilation is allowed its own positive scalar rescaling of time and momenta. In the paper’s C_0^1 -RKHS class, the only exactly covariant kernel is the zero kernel. Conversely, every constant positive-semidefinite matrix kernel has exact dilation covariance, so the classification is tight; these and only these examples are degenerate.

The source conclusion states a broader program rather than a formal conjecture, so this is classified as a substantial partial result. It settles the fixed stationary-kernel version of exact scale covariance, but not multiscale families, approximate invariance, or the other geometric questions in the conclusion.

Source Direction

Micheli and Glaunès develop matrix-valued reproducing kernels for landmark and diffeomorphic shape deformation. For a translation-invariant kernel $K(x, y) = \kappa(x - y)$, their landmark Hamiltonian is

$$H(p, q) = \frac{1}{2} \sum_{a,b=1}^N p_a^\top \kappa(q^a - q^b) p_b,$$

and the first Hamilton equation is

$$\dot{q}^a = \sum_{b=1}^N \kappa(q^a - q^b) p_b. \tag{1}$$

Their conclusion observes that all kernels considered there contain a length constant and that the resulting landmark and diffeomorphism dynamics are not scale invariant. It then identifies application-adapted kernel design and the relation between kernels and geodesic shape properties as future work:

Figure 12: (A) Detail of the exponential map and (B) a landmark trajectory in the divergence-free case.

6 Conclusions

In this paper we have provided a thorough and systematic study of matrix-valued definite positive kernels that induce translation- and rotation-invariant metrics in the corresponding Reproducing Kernel Hilbert Spaces. We have conducted a novel analysis of such kernels in the Fourier domain: here the classification of spaces of curl-free and divergence free vector fields is very natural, in that each corresponds to one of the two coefficients of the Fourier transform of the kernel being equal to zero. This allows one to perform the Hodge decomposition in RKHS via the computation of Hankel transforms (which may be done analytically or numerically). We have also devised a method for building curl-free and divergence-free kernels via an appropriate differentiation of scalar kernels; we have also proven that this constructive procedure is general, in that any TRI kernel (more precisely, the terms of its Hodge decomposition) may be obtained this way. Such method may thus be used to design a large class of positive kernels. We have given a summary of the application of the LDDMM approach to the interpolation of vector fields and the matching of landmark points, together with the differential-geometric interpretation and the derivation of the corresponding Hamiltonian. Finally, we have given numerical examples of the dynamics of landmark points and the corresponding flow of diffeomorphisms for different choices of TRI kernels.

Future work should focus on the design of kernels that are appropriate for different applications, especially in computational anatomy and related fields. For example, the shape of geodesics should be studied in relation to the choice of the kernel; in particular, note that all the kernels that we have considered in the examples have a length constant (e.g. the parameter $\sigma > 0$ for Bessel-type kernels), so that the induced dynamics in the space of landmarks (and, for that matter, in the group of diffeomorphisms itself) is not scale invariant; furthermore, the regularity of the kernels at zero and the heaviness of their tails are also crucial in determining both the qualitative dynamics and the complexity of numerical implementations. Last, but not least, the visualization of the exponential map in the previous section (especially in the case of divergence-free TRI kernels) reveals the presence of several conjugate points, i.e. landmark configurations that are connected by distinct geodesics; the study of the dynamics of $\mathcal{L}^N(\mathbb{R}^d)$ induced by the action of diffeomorphism groups should be furthered, and the investigation of the differential geometry of the space of landmarks that was started in [38] in the case of scalar kernels, which is useful for the ultimate goal of performing statistics on such manifold, should be extended to the case of generic TRI kernels.

Definition and Proof Intuition

Fix the two-landmark system. We call it *exactly dilation covariant* if for every $\lambda > 0$ there are numbers $A_\lambda, C_\lambda > 0$ such that every solution (q, p) is transformed into another solution of the same, fixed-kernel Hamiltonian system by

$$Q^a(t) = \lambda q^a(C_\lambda t), \quad P_a(t) = A_\lambda p_a(C_\lambda t). \quad (2)$$

This definition is deliberately permissive: both time and all momenta may be renormalized by arbitrary dilation-dependent scalars.

At time zero, choose positions $(x, 0)$ and momenta $(0, v)$. Equation (1) computes the first landmark's velocity using the single cross-kernel value $\kappa(x)v$. Applying the same equation to the dilated trajectory compares this with $\kappa(\lambda x)v$, forcing

$$\kappa(\lambda x) = h_\lambda \kappa(x) \quad \text{for all } x, \quad h_\lambda = \frac{\lambda C_\lambda}{A_\lambda}.$$

Positive definiteness prevents a nonzero stationary kernel from having $\kappa(0) = 0$. Evaluation at $x = 0$ therefore forces $h_\lambda = 1$. Sending λ to zero and using continuity then makes κ constant. Constant kernels move every landmark through the same finite-dimensional velocity subspace and have singular multi-landmark Gram matrices.

Main Obstruction

We use the standard matrix-valued positive-definiteness convention:

$$\sum_{i,j=1}^m u_i^\top \kappa(x_i - x_j) u_j \geq 0$$

for all finite choices of $x_i, u_i \in \mathbb{R}^d$. It implies $\kappa(-x) = \kappa(x)^\top$ and $\kappa(0)$ is positive semidefinite.

Lemma 1 (A stationary positive-definite kernel cannot vanish at zero). *If $\kappa : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ is positive definite and $\kappa(0) = 0$, then $\kappa \equiv 0$.*

Proof. For $x, u, v \in \mathbb{R}^d$, positive semidefiniteness of the scalar 2×2 compression of the block Gram matrix at x and 0 gives the kernel Cauchy–Schwarz inequality

$$\left| u^\top \kappa(x) v \right|^2 \leq (u^\top \kappa(0) u) (v^\top \kappa(0) v).$$

If $\kappa(0) = 0$, the right side vanishes for every u, v . Thus every matrix coefficient of $\kappa(x)$ is zero, for every x . $\square$

Theorem 1 (Exact dilation covariance obstruction). *Let $K(x, y) = \kappa(x - y)$ be a nonzero, continuously differentiable, translation-invariant positive-definite matrix-valued kernel on $\mathbb{R}^d$. Suppose the two-landmark Hamiltonian system associated to K is exactly dilation covariant in the sense of (2). Then*

$$\kappa(x) = \kappa(0) \quad (x \in \mathbb{R}^d).$$

Consequently, for every $N \geq 2$, the $Nd \times Nd$ landmark block Gram matrix is singular. In particular, K cannot induce a nondegenerate Riemannian metric on the space of N distinct landmarks. Conversely, every constant positive-semidefinite matrix kernel is exactly dilation covariant.

Proof. Fix $\lambda > 0$. Take arbitrary $x, v \in \mathbb{R}^d$ and a solution with initial data

$$q^1(0) = x, \quad q^2(0) = 0, \quad p_1(0) = 0, \quad p_2(0) = v.$$

The first Hamilton equation (1) gives

$$\dot{q}^1(0) = \kappa(x)v.$$

Differentiate the transformed position in (2):

$$\dot{Q}^1(0) = \lambda C_\lambda \kappa(x)v.$$

On the other hand, because (Q, P) is a solution of the same system, its initial data and (1) give

$$\dot{Q}^1(0) = \kappa(Q^1(0) - Q^2(0))P_2(0) = A_\lambda \kappa(\lambda x)v.$$

Since v is arbitrary,

$$\kappa(\lambda x) = h_\lambda \kappa(x) \quad \text{for all } x \in \mathbb{R}^d, \quad h_\lambda := \frac{\lambda C_\lambda}{A_\lambda} > 0. \quad (3)$$

Lemma 1 and the assumption $K \neq 0$ imply $\kappa(0) \neq 0$. Setting $x = 0$ in (3) yields $(1 - h_\lambda)\kappa(0) = 0$, hence $h_\lambda = 1$. Therefore

$$\kappa(\lambda x) = \kappa(x) \quad (\lambda > 0, x \in \mathbb{R}^d).$$

For fixed x , let $\lambda \downarrow 0$. Continuity at the origin gives $\kappa(x) = \kappa(0)$.

For an N -landmark configuration, the block Gram matrix is now

$$G_N = (\mathbf{1}_N \mathbf{1}_N^\top) \otimes \kappa(0).$$

Its rank is at most d , whereas its size is Nd . It is therefore singular for every $N \geq 2$. The landmark metric tensor in the source paper is the inverse of this block Gram matrix, so no nondegenerate metric results.

For the converse, let $\kappa(x) = B$, where B is positive semidefinite. Then the Hamiltonian equations reduce to

$$\dot{q}^a = B \sum_{b=1}^N p_b, \quad \dot{p}_a = 0.$$

For every $\lambda > 0$, taking $C_\lambda = 1$ and $A_\lambda = \lambda$ shows directly that $Q(t) = \lambda q(t)$, $P(t) = \lambda p(t)$ is again a solution. $\square$

Corollary 1 (The C_0^1 -RKHS class has only the zero kernel). *Assume, as in the source paper, that K is the reproducing kernel of a Hilbert space V continuously embedded in $C_0^1(\mathbb{R}^d, \mathbb{R}^d)$. If the fixed-kernel landmark dynamics are exactly dilation covariant, then $K = 0$.*

Proof. If $K \neq 0$, Theorem 1 makes $\kappa(x) = \kappa(0)$. For every $y, a \in \mathbb{R}^d$, the kernel section

$$x \mapsto K(x, y)a = \kappa(0)a$$

belongs to V , hence lies in C_0^1 and must vanish at infinity. A constant function with that property is zero. Thus $\kappa(0)a = 0$ for every a , contradicting $K \neq 0$. $\square$

What the Result Says About Kernel Design

The sharp classification shows that the obstruction is caused by the conjunction of four requirements: a single fixed kernel, stationarity, ordinary positive definiteness with a finite value at the origin, and exact covariance under all dilations. It is not merely a defect of Gaussian or Bessel examples. Any kernel in this class would have to be homogeneous of degree zero; regularity at the origin collapses degree-zero homogeneity to constancy, and constancy collapses the landmark metric.

Thus exact scale robustness requires changing at least one ingredient. Viable directions include a family κ_σ with a length parameter transformed as $\sigma \mapsto \lambda\sigma$, a dynamical scale variable, a configuration-dependent normalization, or a singular/conditionally positive-definite construction outside the source's RKHS assumptions. The theorem does not distinguish which enlargement is most appropriate for an application.

Remark 1 (Why a scale family escapes). *If $\kappa_\sigma(x) = \phi(x/\sigma)$, then $\kappa_{\lambda\sigma}(\lambda x) = \kappa_\sigma(x)$. This is covariance of a family: the dilation changes the kernel parameter. Theorem 1 concerns covariance of the dynamics generated by one fixed kernel, exactly the setting in which a built-in length constant breaks scale invariance.*

Verification

The necessary direction is analytic and uses only the first Hamilton equation; the converse follows by writing the constant-kernel equations explicitly. The accompanying dependency-free Python script performs three deterministic sanity checks:

1. exact rational row reduction confirms that $(\mathbf{1}_N \mathbf{1}_N^T) \otimes \kappa(0)$ has rank 2, not $2N$, for a positive-definite 2×2 choice of $\kappa(0)$ and $N = 2, 3, 4$;
2. a Gaussian example has dilation ratio 1 at the origin but e^{-3} at unit distance when $\lambda = 2$, so it cannot satisfy (3) with one scalar h_λ ;
3. a two-point scalar block with zero diagonal and nonzero cross term has quadratic values of both signs, illustrating Lemma 1.

The exact command and captured output are recorded in `verification.md`.

Limitations and Novelty Check

This packet does not solve the source’s full program of designing application-specific kernels or relating their tails and regularity to conjugate points and generic geodesic geometry. It also does not rule out approximate scale invariance, invariance over a bounded range, or the model enlargements listed above.

Before packaging, the run’s cheap title, identifier, and keyword indexes were checked for earlier treatment of arXiv:1308.5739. Bounded web searches on 2026-07-22 for the exact title and combinations of *LDDMM*, *landmark Hamiltonian*, *scale invariant*, *dilation covariance*, and *homogeneous positive-definite kernel* found the source paper and general kernel or computer-vision material, but no prior statement of this fixed-kernel obstruction. This is evidence of novelty, not an exhaustive literature review.

Human Review Recommendation

Review as a likely valid substantial partial result. The critical points are the quantifiers in the covariance definition, the two-landmark velocity comparison leading to (3), the use of positive definiteness at the origin, and the distinction between a fixed kernel and a scale-indexed family. The argument requires no claim about the second Hamilton equation beyond existence of the flow and covariance of the full solution.

References

- [1] M. Micheli and J. A. Glaunès, *Matrix-valued Kernels for Shape Deformation Analysis*, arXiv:1308.5739, 2013.